MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, STRAIN RELIEF, 90°, SELF-LOCKING AND NONSELF-LOCKING, CATEGORY 4B (FOR MIL-C-38999 SERIES I AND II CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the accessories described herein shall consist of this specification and the latest issue of MIL-C-85049

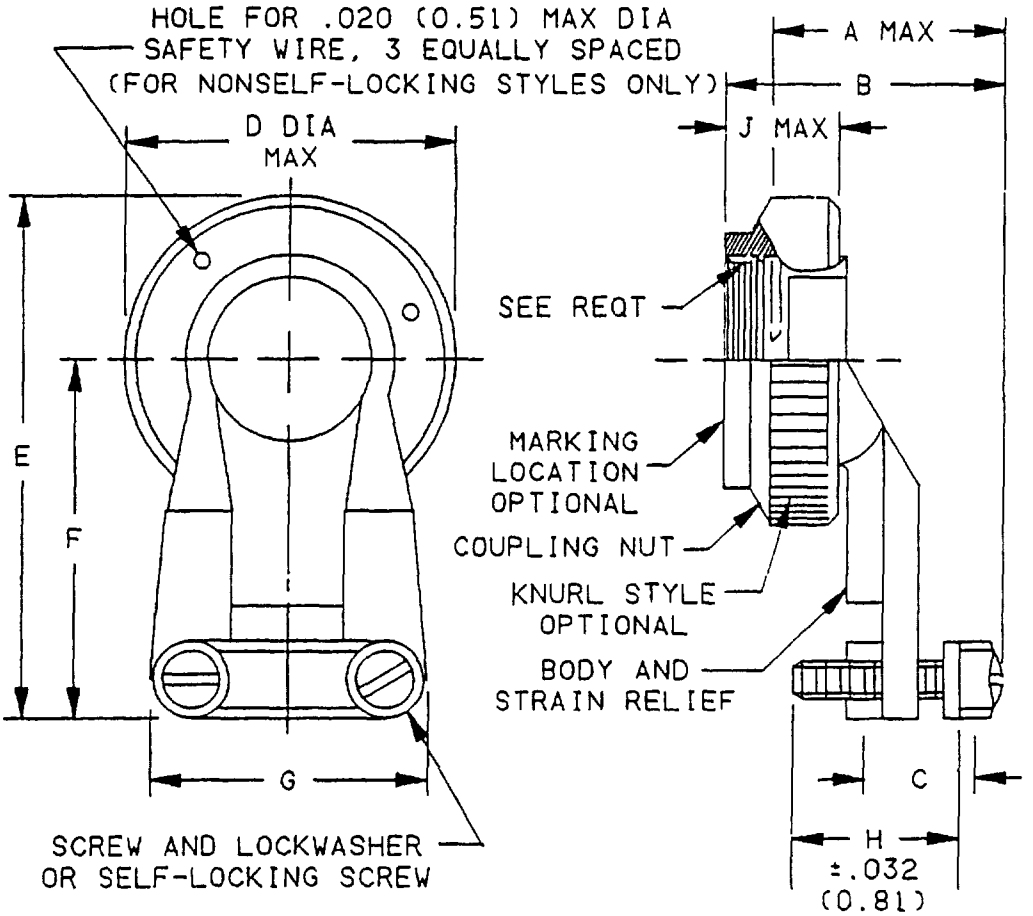


FIGURE 1 Dimensions and configuration (self-locking and nonself-locking)

(B) denotes changes

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Shell sizes and dimensions

Dash no	Connector shell size		A max	B max open	C		D max		E max	F max	G max	H Screw length	Screw size	J max	
	Series II	Series I			Max	Min	Self-locking	Non-self-locking						Self-locking	Non-self-locking
8	8	9	1.16 (29 5)	1.38 (35 1)	.234 (5.94)	.098 (2 49)	86 (21.8)	.75 (19 1)	1 38 (35 1)	1 00 (25.4)	.85 (21.6)	.500 (12.70)	6-32	590 (14.99)	.540 (13.72)
10	10	11	1 16 (29 5)	1.38 (35 1)	234 (5.94)	.153 (3 89)	.98 (24 9)	.85 (21 6)	1.53 (38.9)	1.10 (27 9)	90 (22.9)	.500 (12.70)	6-32	.590 (14 99)	.540 (13.72)
12	12	13	1 24 (31 5)	1.46 (37.1)	328 (8 33)	.190 (4.83)	1 16 (29 5)	1 00 (25.4)	1.60 (40 6)	1.10 (27.9)	1 10 (27.9)	.625 (15 88)	6-32	590 (14 99)	.540 (13.72)
14	14	15	1 41 (35 8)	1.63 (41 4)	.457 (11 61)	.260 (6 60)	1 28 (32 5)	1 10 (27.9)	1.80 (45.7)	1 25 (31 8)	1.15 (29 2)	.750 (19.05)	6-32	.590 (14 99)	.540 (13 72)
16	16	17	1 58 (40.1)	1 80 (45.7)	634 (16 10)	283 (7 19)	1.41 (35.8)	1 25 (31 8)	1 93 (49 0)	1.30 (33.0)	1 30 (33.0)	.750 (19 05)	6-32	590 (14.99)	.540 (13.72)
18	18	19	1 60 (40 6)	1.82 (46.2)	.614 (15 60)	.325 (8 26)	1.52 (38 6)	1.40 (35.6)	2.05 (52.1)	1 35 (34 3)	1 50 (38.1)	.750 (19 05)	8-32	590 (14.99)	.540 (13.72)
20	20	21	1 68 (42.7)	1.90 (48.3)	698 (17.73)	.343 (8 71)	1 64 (41 7)	1 50 (38 1)	2.35 (59 7)	1.60 (40 6)	1.60 (40 6)	875 (22.23)	8-32	.590 (14 99)	.540 (13.72)
22	22	23	1 82 (46 2)	2.04 (51 1)	823 (20 90)	381 (9 68)	1.77 (45.0)	1 65 (41 9)	2.58 (65 5)	1.75 (44 5)	1.70 (43.2)	1 000 (25 40)	8-32	590 (14 99)	.540 (13.72)
24	24	25	1 93 (49 0)	2.15 (54.6)	.853 (21.67)	.418 (10 62)	1 89 (48.0)	1 75 (44 5)	2.73 (69 3)	1.85 (47 0)	1.80 (45.7)	1 125 (28 58)	8-32	590 (14 99)	.540 (13 72)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3 Metric equivalents are given for general information only.
4. Dimensions apply after plating.
5. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configuration (self-locking and nonself-locking) - Continued

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REQUIREMENTS

Self-locking and nonself-locking styles:

Design and construction:

Dimensions and configurations: See figure 1.

Interface dimensions In accordance with MIL-C-38999, series I and II

Accessory: Consists of a coupling nut and body. The coupling nut shall be captivated to, and free to rotate on the body. This accessory will not accommodate connectors using size 8, 4, or 0 contacts.

Material and finish: See table I. Accessories shall be as shown in table I

Clamp screws and lockwashers: 300 series corrosion-resistant steel, passivate.

TABLE I. Material and finish.

Figure	Material	Finish
1	Aluminum alloy in accordance with MIL-C-85049	W N 1/
1	Corrosion-resistant steel, 300 series in accordance with QQ-S-763	S

1/ Not for Navy use. Air Force use for space applications only.

Self-locking styles only:

Self-locking device within the coupling nut shall be corrosion-resistant metal and shall provide a positive detent. Couplings with self-locking devices shall meet all the performance requirements specified herein for its specific category and the following additional tests.

Life cycle: The coupling with the locking device engaged shall be rotated 60 full turns in a clockwise direction and 60 full turns counter clockwise, prior to standard qualification testing Cycle speed shall not exceed one full turn per second.

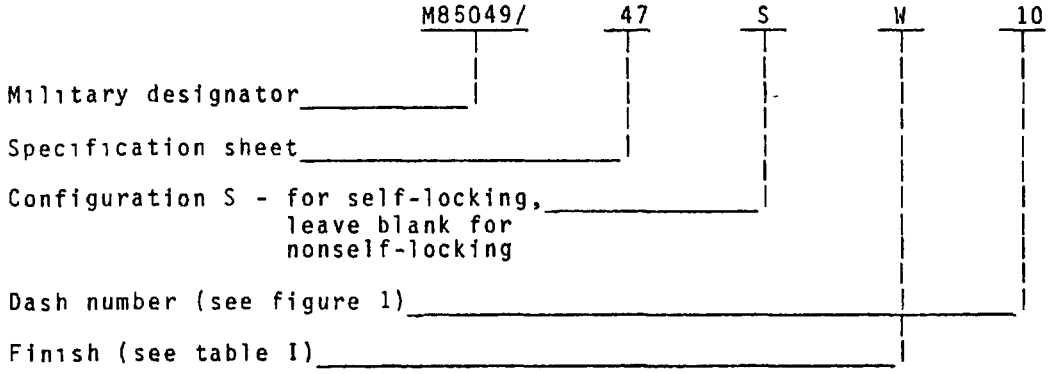
Locking: Coupling nut torque shall be 80 percent of coupling thread strength values given for medium and light duty accessories as specified in table III of MIL-C-85049 In table III where N/A appears, the values shall be 80 percent of those values used in qualification testing for coupling thread strength Tolerance shall be ±5 inch-pounds.

Vibration Backshells shall be subjected to test parameters of applicable connector specifications. The coupling torque shall be within +20, -10 inch-pounds of the initial value.

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Part number. See example below:

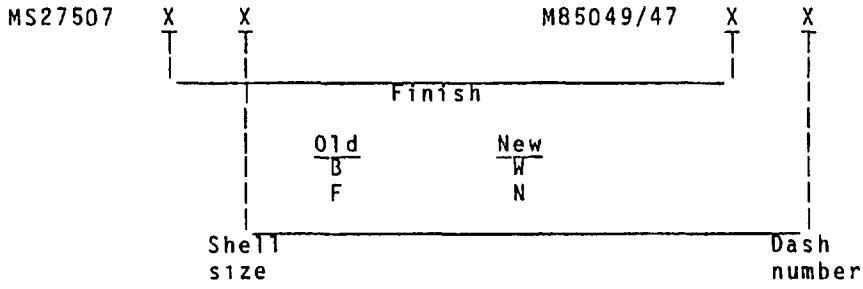
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Qualification: See MIL-C-85049, category 4B. Accessory shall be tested with MIL-C-38999 series I connectors.

QPL evaluating activity: Naval Avionics Center, Code B/714, 6000 East 21st Street, Indianapolis, IN 46219-2189.

Supersession data:



CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC
Air Force - 11, 15, 19, 99
DLA - ES

Ⓑ Civil Agencies Coordinating Activities:
NASA - NA

Preparing activity:
Air Force - 85

Agent.
DLA - ES

(Project 5935-3677)

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